Sequences and Series

Questions 1–200

2.06 — Question 1. For each of the following sequences, explain whether they can be described as convergent, divergent, periodic, or none of these.

a) $1, 2, 4, 8, 16, 32, \ldots$

b) $1, 5, 1, 5, 1, 5, \ldots$

c) $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \ldots$

d) $50, 60, 65, 67.5, 68.75, 69.375, \ldots$

2.06 — Question 2. Sigma Notation

Evaluate each of these:

a) $\displaystyle\sum_{n=1}^{4}(5n-1)$

b) $\displaystyle\sum_{n=1}^{3}(3^n+n^2)$

c) $\displaystyle\sum_{n=1}^{5}\frac{2}{n}$

d) $\displaystyle\sum_{n=1}^{4}4\times\left(\frac{1}{2}\right)^{n-1}$

2.06 — Question 3. Evaluate each of these:

a) $\sum_{n=3}^{6}(n^2 - 10)$

b) $\sum_{n=99}^{100}\frac{3}{n}$

c) $\sum_{n=6}^{8}50 \times 5^{n-5}$

2.06 — Question 4. Simplify each of these, where k is a constant:

a) $\sum_{n=1}^{5}(n + k)$

b) $\sum_{n=1}^{4}(1 - kn^3)$

c) $\sum_{n=2}^{4}\frac{k}{n+1}$

d) $\sum_{n=3}^{6}\frac{kn}{k+1}$

2.06 — Question 5. It is given that $f(n)$ is the nth term of a sequence. Some further information about $f(n)$ is given in the table below.

k	19	20	21	22	23	24	25
$\sum_{n=1}^{k} f(n)$	$\frac{19}{10}$	$\frac{21}{10}$	$\frac{58}{25}$	$\frac{127}{50}$	$\frac{69}{25}$	3	$\frac{163}{50}$

Find the value of

$$\sum_{n=20}^{24} f(n)$$

2.06 — Question 6. It is given that

$$g(k) = \sum_{n=1}^{k} (n+2)^2$$

where k is an integer such that $k \geq 1$. Find:

a) $g(2)$
b) $g(5)$
c) $g(7) - g(6)$
d) $g(k+1) - g(k)$

2.06 — Question 7. Find

$$\sum_{i=1}^{100} 1 + \sum_{i=20}^{100} 1 + \sum_{i=50}^{80} 1$$

2.06 — Question 8. Show that

$$\sum_{n=1}^{4} \frac{6(n+1)}{\sqrt{n}} = 27 + 9\sqrt{2} + 8\sqrt{3}$$

2.06 — Question 9. Show that

$$\sum_{n=1}^{3} \frac{n}{\sqrt{n}+\sqrt{n+1}} = 5 - \sqrt{2} - \sqrt{3}$$

2.06 — Question 10. Show that

$$\sum_{n=1}^{3} \ln(n+1) = \ln 24$$

2.06 — Question 11. Show that

$$\sum_{n=1}^{4} \ln\left(\frac{1}{n}\right) = -\ln 24$$

2.06 — Question 12. Show that

$$\sum_{n=1}^{5} \ln\left(\frac{n+1}{n}\right) = \ln 6$$

2.06 — Question 13. It is given that

$$h(k) = \sum_{n=1}^{k} \frac{n}{n+1}$$

Find the value of k such that

$$h(k+1) - h(k) = \frac{99}{100}$$

2.06 — Question 14. Recurrence Relations and Limits

It is given that

$$u_{n+1} = 3u_n + 4$$

with $u_1 = 50$. Find the values of u_2, u_3 and u_4.

2.06 — Question 15. It is given that

$$u_{n+1} = \frac{u_n + 2}{10}$$

with $u_1 = 1000$. Find the values of u_2, u_3 and u_4.

2.06 — Question 16. It is given that

$$u_{n+1} = 5u_n + k$$

where k is a constant, with $u_1 = 2$ and $u_3 = 4$. Find the value of k.

2.06 — Question 17. It is given that

$$u_{n+1} = \frac{1}{u_n + 2}$$

with $u_1 = 3$. Calculate

$$\sum_{n=1}^{3} u_n$$

2.06 — Question 18. It is given that

$$u_{n+1} = \frac{10}{u_n} - k$$

where k is a constant, with $u_1 = 10$. It is also given that

$$\sum_{n=1}^{3} u_n = 0$$

Find the two possible values of k.

2.06 — Question 19. In each of the following cases, use your calculator to determine if the sequence converges or diverges.

a) $u_{n+1} = 0.3u_n - 8,\ u_1 = 1$
b) $u_{n+1} = 1.1u_n - 3,\ u_1 = 2$
c) $u_{n+1} = 2.5u_n + 2,\ u_1 = -3$
d) $u_{n+1} = 3 - 1.8u_n,\ u_1 = 4$
e) $u_{n+1} = 100 - 0.9u_n,\ u_1 = 10$

2.06 — Question 20. In each of the following cases, determine if the sequence converges or diverges.

a) $u_n = 3 + 0.6n$
b) $u_n = 2^n$
c) $u_n = \left(\frac{1}{3}\right)^n$
d) $u_n = \frac{10}{n+1}$

2.06 — Question 21. In each of the following cases, it is given that the limit of u_n as n tends to infinity is L. In each case, write down an equation for L and find its value.

a) $u_{n+1} = \frac{u_n}{3} + 50$
b) $u_{n+1} = 0.85u_n - 6$
c) $u_{n+1} = \frac{u_n + 15}{30}$
d) $u_{n+1} = \frac{0.2u_n - 100}{25}$

2.06 — Question 22. Arnold takes an initial dose of 400 mg of a certain type of painkiller, and every four hours must take another dose. The amount of painkiller in Arnold's system at the end of a four-hour period, just after he has taken the next dose, can be modelled by the recurrence relation $P_{n+1} = 0.25P_n + 400$.

a) Explain the significance of 0.25 in the recurrence relation.
b) Show that the amount of painkiller in Arnold's system 12 hours after he took the first dose, is 531 mg to the nearest mg.
c) Use algebra to determine the long-term amount of painkiller in Arnold's system.

2.06 — Question 23. In each of the following cases, determine the possible value(s) of u_1 that will generate a constant sequence.

a) $u_{n+1} = \frac{9u_n}{8} + 5$
b) $u_{n+1} = \frac{10-u_n}{15}$
c) $u_{n+1} = \frac{5}{2} + \frac{21}{u_n}$
d) $u_{n+1} = \frac{55-3u_n-8u_n^2}{10u_n}$

2.06 — Question 24. Write down the limit of the following sequences:

a) $u_n = \frac{1}{n+10}$
b) $u_n = \frac{n}{n+10}$
c) $u_n = \frac{n+11}{n+10}$
d) $u_n = \frac{2n+11}{n+10}$
e) $u_n = \frac{2n+11}{n^2+10}$

2.06 — Question 25. It is given that $u_{n+1} = 1 - u_n$ with $u_1 = 1$. Find the value of u_{155}.

2.06 — Question 26. It is given that $u_{n+1} = 8 - u_n$ with $u_1 = 1$. Find the value of u_{88}.

2.06 — Question 27. It is given that $u_{n+1} = -u_n$ with $u_1 = 1$.

a) Find the value of u_{100}

b) Find the value of $\sum_{n=1}^{100} u_n$

2.06 — Question 28. It is given that $u_{n+1} = 13 - u_n^2$ with $u_1 = 4$.

a) Find the value of u_{500}

b) Find the value of $\sum_{n=1}^{500} u_n$

2.06 — Question 29. It is given that $u_{n+1} = -\frac{1}{u_n}$ with $u_1 = \frac{5}{3}$.

a) Find the value of u_{99}

b) Find the value of $\sum_{n=1}^{99} u_n$

2.06 — Question 30. It is given that

$$u_{n+1} = \frac{u_n - 1}{k u_n}$$

with $u_1 = 3$, where k is a constant, generates a sequence with period 3.

a) Find the value of k

b) Hence find the value of $\sum_{n=1}^{100} u_n$

2.06 — Question 31. It is given that $u_{n+3} = u_n$ with $u_1 = 4$, $u_2 = -6$ and $u_3 = 8$. Find the value of

$$\sum_{n=1}^{500} u_n$$

2.06 — Question 32. It is given that $u_{n+4} = u_n$ with $u_1 = 1$, $u_2 = 4$, $u_3 = -7$ and $u_4 = -2$. Find the value of

$$\sum_{n=1}^{499} u_n$$

2.06 — Question 33. It is given that $u_{n+3} = u_n + 2$ with $u_1 = 0$, $u_2 = 1$ and $u_3 = 5$. Find k such that $u_k = 200$.

2.06 — Question 34. Periodic, Increasing, Decreasing
It is given that

$$u_n = 3 + (-1)^n$$

is a periodic sequence. State the period of the sequence.

2.06 — Question 35. It is given that

$$u_n = \sin\left(\frac{3\pi n}{2}\right)$$

is a periodic sequence. State the period of the sequence.

2.06 — Question 36. It is given that

$$u_n = \cos\left(\frac{2\pi n}{3}\right)$$

is a periodic sequence. State the period of the sequence.

2.06 — Question 37. It is given that

$$u_n = \sin\left(\frac{2\pi n}{3}\right) - \cos\left(\frac{5\pi n}{3}\right)$$

is a periodic sequence. State the period of the sequence.

2.06 — Question 38. It is given that

$$u_n = \cos\left(\frac{\pi n}{4}\right) - \sin\left(\frac{5\pi n}{2}\right)$$

is a periodic sequence. State the period of the sequence.

2.06 — Question 39. It is given that

$$u_n = (-1)^n + \sin\left(\frac{\pi n}{2}\right)$$

is a periodic sequence. State the period of the sequence.

2.06 — Question 40. It is given that

$$u_n = 3(-1)^n + \cos\left(\frac{\pi n}{3}\right)$$

is a periodic sequence. State the period of the sequence.

2.06 — Question 41. It is given that

$$u_n = 2(-1)^n - \frac{1}{\cos(\pi n/3)}$$

is a periodic sequence. State the period of the sequence.

2.06 — Question 42. Show that

$$u_n = \frac{3n}{2+n^2}$$

is not an increasing sequence.

2.06 — Question 43. Show that

$$u_n = \frac{n+1}{n^3+n-1}$$

is not an increasing sequence.

2.06 — Question 44. Show that

$$u_n = n - \frac{1}{n+1}$$

is not a decreasing sequence.

2.06 — Question 45. Show that

$$u_n = n^2 - 100n$$

is not a decreasing sequence.

2.06 — Question 46. Prove that

$$u_n = 3^{-n}$$

is a decreasing sequence.

2.06 — Question 47. Prove that

$$u_n = \frac{1}{n^2}$$

is a decreasing sequence.

2.06 — Question 48. Prove that

$$u_n = \frac{n+1}{n+2}$$

is an increasing sequence.

2.06 — Question 49. Prove that

$$u_n = \frac{n+3}{n+2}$$

is a decreasing sequence.

2.06 — Question 50. Prove that

$$u_n = \frac{5}{n+5}$$

is a decreasing sequence.

2.06 — Question 51. Prove that

$$u_n = \frac{2-n}{n}$$

is a decreasing sequence.

2.06 — Question 52. Prove that

$$u_n = (2n+1)^2 - (n-1)^2$$

is an increasing sequence.

2.06 — Question 53. Prove that

$$u_n = \sqrt{n+1} + \sqrt{n}$$

is an increasing sequence.

2.07 — Question 1. Write the nth term, u_n, for each of these arithmetic sequences in the form

$$u_n = a + (n-1)d$$

where a is the first term and d is the common difference.

a) $3, 8, 13, 18, 23, \ldots$

b) $25, 35, 45, 55, 65, \ldots$

c) $12, 6, 0, -6, -12, \ldots$

d) $150, 135, 120, 105, 90, \ldots$

e) $-44, -41, -38, -35, -32, \ldots$

f) $-100, -121, -142, -163, -184, \ldots$

g) $\frac{5}{6}, \frac{31}{30}, \frac{37}{30}, \frac{43}{30}, \frac{49}{30}, \ldots$

h) $\frac{11}{4}, -\frac{7}{4}, -\frac{25}{4}, -\frac{43}{4}, -\frac{61}{4}, \ldots$

2.07 — Question 2. It is given that $9, 12, 15, 18, 21, \ldots$ is an arithmetic sequence. Find the 500th term in the sequence.

2.07 — Question 3. It is given that $\frac{3}{2}, \frac{11}{6}, \frac{13}{6}, \frac{5}{2}, \frac{17}{6}, \ldots$ is an arithmetic sequence. Find the 240th term in the sequence.

2.07 — Question 4. It is given that $-50, -45, -40, -35, \ldots, 765$ is a finite arithmetic sequence. Determine how many terms there are in the sequence.

2.07 — Question 5. It is given that $31.3, 31.6, 31.9, 32.2, \ldots, 117.7$ is a finite arithmetic sequence. Determine how many terms there are in the sequence.

2.07 — Question 6. It is given that $58.2, 57.5, 56.8, 56.1, \ldots, -25.1$ is a finite arithmetic sequence. Determine how many terms there are in the sequence.

2.07 — Question 7. An arithmetic progression has 3rd term 22 and 7th term 54. Find the first term and the common difference.

2.07 — Question 8. An arithmetic progression has 5th term 38 and 10th term 50.5. Find the first term and the common difference.

2.07 — Question 9. An arithmetic progression has 2nd term 71 and 9th term 631. Find the first term and the common difference.

2.07 — Question 10. An arithmetic progression has 6th term 20 and 12th term 38. Find the first term that exceeds 100.

2.07 — Question 11. An arithmetic sequence has 3rd term $2+\sqrt{3}$ and 9th term $3+\sqrt{5}$. Find the common difference in the form $a+b\sqrt{3}+c\sqrt{5}$, where a, b and c are rational numbers.

2.07 — Question 12. A botanist is studying the growth of a particular species of plant. The height of the plant after 2 weeks is 15 cm, and after 7 weeks it is 55 cm. Given the height of the plant over the weeks follows an arithmetic sequence, find the height of the plant after 3 weeks.

2.07 — Question 13. An arithmetic sequence has the three consecutive terms $\ln \frac{1}{4}k$, $\ln \frac{1}{3}k$, $\ln p$, where k, p are positive constants. Find p in terms of k.

2.07 — Question 14. An arithmetic sequence has the three consecutive terms $6k^2$, $5k$, $(21k - 35)$, where k is a constant. Find the possible values of k.

2.07 — Question 15. An arithmetic sequence has the three consecutive terms $\frac{2}{k}$, $20k$, $-(5k + 13)$, where k is a constant. Find the possible values of k.

2.07 — Question 16. An arithmetic sequence has the three consecutive terms $16e^{k}$, 20, $16e^{-k}$, where k is a constant. Find the possible values of k.

2.07 — Question 17. An arithmetic sequence has the three consecutive terms 28, $\frac{200}{2^k}$, $\left(\frac{240}{2^k} - 2^k\right)$, where k is a constant. Find the possible values of k.

2.07 — Question 18. The first three terms of an arithmetic sequence are $32 + 3k$, $13 - 4k$, $78 + 10k$, where k is a constant.

a) Find the value of k.

b) Find the 100th term in the sequence.

2.07 — Question 19. Find the sum of the arithmetic series $3 + 11 + 19 + 27 + \cdots + 155$.

2.07 — Question 20. Find the sum of the arithmetic series $14 + 20 + 26 + 32 + \cdots + 1238$.

2.07 — Question 21. Find the sum of the arithmetic series $-11 - 20 - 29 - 38 - \cdots - 542$.

2.07 — Question 22. Find the sum of the arithmetic series $\frac{20}{3} + \frac{35}{6} + 5 + \frac{25}{6} + \cdots - 60$.

2.07 — Question 23. An arithmetic series is given by $\sum_{i=7}^{40} \left(6i - \frac{1}{2}\right)$.

a) Write down the first term and common difference of the series.

b) Write down how many terms there are in the series.

2.07 — Question 24. An arithmetic sequence has rth term given by $u_r = 7 - 10r$. Show that $\sum_{r=1}^{40} u_r = -7920$.

2.07 — Question 25. An arithmetic sequence has kth term given by $u_k = -3 + \frac{k}{3}$. Show that $\sum_{k=10}^{30} u_k = 77$.

2.07 — Question 26. Find the sum of the first 150 terms of the arithmetic sequence $25 + 26.5 + 28 + 29.5 + \cdots$.

2.07 — Question 27. Find the sum of the first 80 terms of the arithmetic sequence $\frac{1}{5} + \frac{7}{5} + \frac{13}{5} + \frac{19}{5} + \cdots$.

2.07 — Question 28. Find the sum of the first 185 terms of the arithmetic sequence $100 + 99.2 + 98.4 + 97.6 + \cdots$.

2.07 — Question 29. Find the sum of the first 390 terms of the arithmetic sequence $10.3 + 10 + 9.7 + 9.4 + \cdots$.

2.07 — Question 30. On his 1st birthday, Jaime's parents put £35 into a bank account. On his 2nd birthday, Jaime's parents put £65 into the bank account. The amounts deposited into the bank account form an arithmetic sequence. Determine the total amount of money in the bank account the day after Jaime's 18th birthday.

2.07 — Question 31. It is given that S_n is the sum of the first n terms of the arithmetic sequence $104 + 101 + 98 + 95 + \cdots$. Find the maximum value of S_n.

2.07 — Question 32. An arithmetic sequence has nth term given by $u_n = \frac{6n}{5} + \frac{2}{3}$. Find the value of k such that

$$\sum_{n=1}^{k} u_n < 200 \quad \text{and} \quad \sum_{n=1}^{k+1} u_n > 200.$$

2.07 — Question 33. An arithmetic sequence has nth term given by $u_n = 4.1n - 3$. Find the value of k such that

$$\sum_{n=1}^{k} u_n < 3000 \quad \text{and} \quad \sum_{n=1}^{k+1} u_n > 3000.$$

2.07 — Question 34. Aiden buys a new phone on a contract that is worth £900 at 0% interest. The contract provider requires Aiden to pay back £10.50 at the end of week 1, £10.60 at the end of week 2, £10.70 at the end of week 3, etc, until the £900 is fully paid off, forming an arithmetic sequence. At the end of the final week, Aiden pays off the remaining balance if it is less than the amount he paid the previous week.

a) Determine how many weeks it will take to pay off the contract.

b) Determine how much Aiden pays on the final week.

2.07 — Question 35. An arithmetic series has 3rd term 46 and 15th term 222. Find the sum of the first 30 terms.

2.07 — Question 36. An arithmetic series has 1st term 3. The sum of the first 100 terms is 12. Find the common difference.

2.07 — Question 37. The first three terms of an arithmetic progression are $8k$, $k^2(k-4)$, 10, where k is a real constant. It is given that the sum of the first n terms is -53900. Find the value of n.

2.07 — Question 38. An arithmetic sequence has first term a and common difference d. The sum of the first 15 terms is -465.

a) Show that $a + 7d = -31$.

It is given that the 50th term is -283.

b) Find the value of a and the value of d.

2.07 — Question 39. An arithmetic sequence has first term a and common difference d. The sum of the first 10 terms is 410.

a) Show that $2a + 9d = 82$.

It is given that the sum of the first 20 terms is 1620.

b) Find the value of a and the value of d.

c) Find the sum of the first 30 terms.

2.07 — Question 40. An arithmetic sequence has first term a and common difference d. The sum of the first 490 terms is 1489 times the sum of the first 10 terms.

a) Show that $3a = 11d$.

b) Given that the 10th term is 38, find the value of a and the value of d.

2.07 — Question 41. An arithmetic progression has first term a and common difference d. The sum of the first 220 terms is 211 times the sum of the first 20 terms.

a) Show that $a + 4d = 0$.

b) Given that the 105th term is -200, find the value of a and the value of d.

2.07 — Question 42. An arithmetic sequence has first term a and common difference d. The sum of the first 425 terms is 221 times the square of the sum of the first 5 terms.

a) Show that $221a^2 - 17a + 884d^2 - 3604d + 884ad = 0$.

b) Given that the 2nd term is 4 and that both a and d are integers, find the value of a and the value of d.

2.07 — Question 43. The first three terms of an arithmetic sequence are given by $\frac{1}{k-4}$, $-\frac{1}{k}$ and $\frac{1}{k+4}$, where k is a positive constant. Show that the sum of the first 52 terms is $a + b\sqrt{2}$, where a and b are integers.

2.07 — Question 44. The first three terms of an arithmetic sequence are 8, 11, and 14. The sum of the first n terms is greater than 1000 but less than 2000. Find the possible values of n.

2.07 — Question 45. An arithmetic series with sum 100 is given by $\sum_{n=1}^{100}(pn+q)$ where p and q are constants. It is also given that $u_{51}+u_{53}+u_{55}=u_{59}$. Find the value of p and the value of q.

2.07 — Question 46. An arithmetic sequence has first term $(3k+2)$ and last term $(9k-1)$, where k is a positive constant. The common difference of the sequence is 6. Find the number of terms in the sequence in terms of k.

2.07 — Question 47. It is given that k is a positive integer. A_k is the sum of the first k terms of an arithmetic series with first term 3 and common difference 8. B_k is the sum of the first k terms of an arithmetic series with first term -52 and common difference 18. Find the value of k such that $A_k = B_k$.

2.07 — Question 48. It is given that k is a positive integer. C_k is the sum of the first k terms of an arithmetic series with first term 1500 and common difference -9. D_k is the sum of the first k terms of an arithmetic series with first term 6 and common difference 3. Find the value of k such that $C_k = D_k$.

2.07 — Question 49. An arithmetic sequence has first term a and common difference d. Show that S_n, the sum of the first n terms, is $\frac{1}{2}n(2a + (n-1)d)$.

2.07 — Question 50. Find the sum of all the even numbers from 100 to 500.

2.07 — Question 51. Find the sum of all the odd numbers from 5 to 555.

2.07 — Question 52. It is given that $u_1 = p$, $u_2 = (2p-1)$, $u_3 = (3p+2)$, $u_4 = (15p+\frac{2}{3})$, ... is **not** an arithmetic sequence, where p is a constant such that $p \geq 1$. However, the sequence formed by the ratios $\frac{u_{n+1}}{u_n}$ is an arithmetic sequence. Find the value of p and hence find the value of u_5.

2.08 — Question 1. Write the nth term, u_n, for each of these geometric sequences in the form

$$u_n = ar^{n-1}$$

where a is the first term and r is the common ratio.

a) $3, 12, 48, 192, 768, \ldots$

b) $80, 120, 180, 270, 405, \ldots$

c) $240, 120, 60, 30, 15, \ldots$

d) $10, 9, 8.1, 7.29, 6.561, \ldots$

e) $22, -44, 88, -176, 352, \ldots$

f) $\frac{1}{3}, -\frac{4}{3}, \frac{16}{3}, -\frac{64}{3}, \frac{256}{3}, \ldots$

g) $\frac{9}{10}, -1, \frac{10}{9}, -\frac{100}{81}, \frac{1000}{729}, \ldots$

h) $88, -66, 49.5, -37.125, 27.84375, \ldots$

2.08 — Question 2. It is given that $1.5, 3, 6, 12, \ldots$ is a geometric sequence. Find the 15th term in the sequence.

2.08 — Question 3. It is given that $6, 8, \frac{32}{3}, \frac{128}{9}, \ldots$ is a geometric sequence. Find the 6th term in the sequence.

2.08 — Question 4. It is given that $0.004, 0.012, 0.036, 0.108, \ldots, 57395.628$ is a finite geometric sequence. Determine how many terms there are in the sequence.

2.08 — Question 5. It is given that $1.02, 4.08, 16.32, 65.28, \ldots, 1069547.52$ is a finite geometric sequence. Determine how many terms there are in the sequence.

2.08 — Question 6. It is given that $10000, 11000, 12100, 13310, \ldots, 21435.8881$ is a finite geometric sequence. Determine how many terms there are in the sequence.

2.08 — Question 7. A geometric progression has first term 0.2 and common ratio 2. Find the first term in the sequence that exceeds 1000.

2.08 — Question 8. A geometric progression has 4th term 28 and 15th term 57344. Find the first term and the common ratio.

2.08 — Question 9. A geometric progression has 3rd term 2 and 7th term 13122. Find the possible first terms and their corresponding common ratios.

2.08 — Question 10. A geometric progression has 2nd term 15 and 5th term -234375. Find the first term and the common ratio.

2.08 — Question 11. A geometric progression has 6th term 62.208 and 7th term 373.248. Find the first term and the common ratio.

2.08 — Question 12. A geometric progression has 3rd term 19.8 and 6th term 534.6. Find the first term in the sequence that exceeds ten million.

2.08 — Question 13. A geometric sequence has 2nd term $1 + 2\sqrt{5}$ and 3rd term $38 + 19\sqrt{5}$.

a) Show that the common ratio is $8 + 3\sqrt{5}$

b) Find the 1st term in the sequence, writing it in the form $p + q\sqrt{5}$, where p and q are rational numbers.

2.08 — Question 14. A geometric sequence has the three consecutive terms $(k+1)$, $5k$, $\frac{50k}{7}$, where k is a constant. Find the value of k.

2.08 — Question 15. A geometric sequence has the three consecutive terms $\frac{49k}{36}$, $(k+2)$, $(3k-2)$, where k is a constant. Find the possible values of k.

2.08 — Question 16. A geometric sequence has the three consecutive terms 2^{3k-7}, 2^{2k+1}, 2^{4k-5}, where k is a constant. Find the value of k.

2.08 — Question 17. A geometric sequence has the three consecutive terms $3e^{7/4+2k}$, $\frac{15}{2}e^{k+67/24}$, $\frac{75}{4}e^{101/24-k}$, where k is a constant. Find the value of k.

2.08 — Question 18. Find the sum of the geometric series

$$5 + 10 + 20 + \cdots + 2560$$

2.08 — Question 19. Find the sum of the geometric series

$$0.13 + 0.52 + 2.08 + \cdots + 2129.92$$

2.08 — Question 20. Find the sum of the geometric series

$$\frac{1}{5000} + \frac{1}{1250} + \frac{2}{625} + \cdots + \frac{8192}{625}$$

2.08 — Question 21. Find the sum of the geometric series

$$9000 - 9900 + 10890 - \cdots - 17538.4539$$

2.08 — Question 22. Find the sum of the geometric series

$$\frac{2}{3} - 1 + \frac{3}{2} - \cdots + \frac{19683}{512}$$

2.08 — Question 23. Aaliyah decides to go on a seven-day walking holiday along a Highland trail, camping at the end of each day. On the first day, she walks 15 miles. On each subsequent day, she walks 8% less than the previous day.

a) Determine how far she walks on the last day, to two decimal places.

b) Determine the total distance Aaliyah walks on her holiday, to two decimal places.

2.08 — Question 24. A geometric series is given by

$$\sum_{i=6}^{25} \frac{4}{3} \times (\frac{2}{5})^{i-1}$$

a) Write down the first term and common ratio of the series.

b) Write down how many terms there are in the series.

2.08 — Question 25. Find the sum of the first 8 terms of the following geometric series to 2 decimal places:

$$66 + 88 + \frac{352}{3} + \cdots$$

2.08 — Question 26. Find the sum of the first 9 terms of the following geometric series to 2 decimal places:

$$5200 + 4160 + 3328 + \cdots$$

2.08 — Question 27. A geometric sequence has kth term given by $u_k = 128 \times (\frac{7}{2})^{k-1}$. Show that

$$\sum_{k=1}^{5} u_k = 26840$$

2.08 — Question 28. A geometric sequence has rth term given by $u_r = 19683 \times (\frac{1}{3})^{r-1}$. Show that

$$\sum_{r=4}^{10} u_r = 1093$$

2.08 — Question 29. A geometric sequence, u_n, has a common ratio $r > 1$. Find the value of r such that

$$\sum_{n=1}^{4} u_n = 8 \sum_{n=1}^{2} u_n$$

2.08 — Question 30. A geometric series has first term 100 and common ratio $\frac{9}{10}$. It is given that the sum of the first k terms is greater than 870.

a) Show that

$$k > \frac{\ln 0.13}{\ln 0.9}$$

b) Hence find the smallest possible value of k

2.08 — Question 31. A geometric series has first term 2 and common ratio $\frac{26}{25}$. It is given that the sum of the first k terms is greater than 1000.

a) Show that

$$k > \frac{\ln 21}{\ln 1.04}$$

b) Hence find the smallest possible value of k

2.08 — Question 32. A geometric series has first term 20 and common ratio $\frac{1}{8}$. It is given that the sum of the first k terms is less than 22.85.

a) Show that

$$k > \frac{\ln 3200}{\ln 8}$$

b) Hence find the smallest possible value of k

2.08 — Question 33. Find the smallest possible value of k such that the sum to k terms of the series

$$5 + 10 + 20 + 40 + \cdots$$

exceeds 1500.

2.08 — Question 34. Find the smallest possible value of k such that the sum to k terms of the series

$$1.1 + 3.3 + 9.9 + 29.7 + \cdots$$

exceeds 30000.

2.08 — Question 35. A small clothes shop made a net profit of £250 in their first week of trading. The shop's owner predicts that the net profit each week will form a geometric sequence, increasing by 3% each week.

a) Find the net profit predicted by the owner for the end of the fifth week.

b) How many weeks will need to have passed since opening for the weekly net profit to be over £1000?

c) Determine the total amount of net profit in one year (52 weeks) of trading?

2.08 — Question 36. Explain why the geometric series

$$7 + 14 + 28 + 56 + \cdots$$

is not convergent and therefore has no finite sum to infinity.

2.08 — Question 37. Determine which of the following geometric series are convergent:

a) $0.4 + 1.2 + 3.6 + 10.8 + \cdots$

b) $10 + 5 + 2.5 + 1.25 + \cdots$

c) $20 + 30 + 45 + 67.5 + \cdots$

d) $100 + 95 + 90.25 + 85.7375 + \cdots$

e) $100 - 95 + 90.25 - 85.7375 + \cdots$

f) $1.6 - 1.44 + 1.296 - 1.1664 + \cdots$

2.08 — Question 38. Find the sum to infinity of the geometric series with the following properties:

a) $a = 30,\ r = 0.1$

b) $a = 800,\ r = \frac{2}{5}$

c) $a = 99,\ r = 0.99$

d) $a = -8000,\ r = \frac{3}{8}$

e) $a = 410,\ r = -\frac{5}{6}$

f) $a = -9,\ r = -\frac{11}{12}$

2.08 — Question 39. A geometric series has first term 80 and sum to infinity 90. Find the common ratio r.

2.08 — Question 40. Find the sum to infinity of the geometric series

$$1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$$

2.08 — Question 41. Find the sum to infinity of the geometric series

$$100 + 80 + 64 + 51.2 + \cdots$$

2.08 — Question 42. Find the sum to infinity of the geometric series

$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{6}} + \frac{1}{3\sqrt{2}} + \frac{1}{3\sqrt{6}} + \cdots$$

2.08 — Question 43. A geometric series has first term 50 and the sum to infinity is less than 100. Find the range of possible values of r.

2.08 — Question 44. A geometric series has first term 9 and the sum to infinity is less than 40. Find the range of possible values of r.

2.08 — Question 45. A geometric series has first term 100 and the sum to infinity is greater than 500. Find the range of possible values of r.

2.08 — Question 46. A geometric series has first term 80 and the sum to infinity is greater than 100 but less than 200. Find the range of possible values of r.

2.08 — Question 47. Find the value of

$$\sum_{n=1}^{\infty} 3000 \times (\frac{2}{3})^{n-1}$$

2.08 — Question 48. Find the value of

$$\sum_{n=3}^{\infty} 25350 \times (\frac{11}{13})^n$$

2.08 — Question 49. Find the value of

$$\sum_{n=10}^{\infty} 1000000 \times (\frac{1}{5})^{n-1}$$

2.08 — Question 50. The geometric sequence u_n has terms $18, \frac{27}{2}, \frac{81}{8}, \frac{243}{32}, \ldots$

a) Find the sum to infinity of the series generated by the terms of u_n

b) The sequence v_n is found by multiplying each term in the sequence u_n by 32. Write down the first four terms of the sequence v_n

c) Using just your answer to a), write down the sum to infinity of the series generated by the terms of v_n

2.08 — Question 51. The 2nd term of a geometric series is 10. The sum to infinity is 40. Find the first term and common ratio.

2.08 — Question 52. The 2nd term of a geometric series is 10800 and the sum to infinity is 45000. Find the two possible first terms of this series.

2.08 — Question 53. The 2nd term of a geometric series is 95 and the sum to infinity is $\frac{9500}{9}$. It is given that $r > \frac{1}{2}$. Find the sum of the first 6 terms in the series to two decimal places.

2.08 — Question 54. A geometric sequence, u_n, has first term a and common ratio r. It is given that

$$\sum_{n=1}^{\infty} u_n = 10 \sum_{n=1}^{2} u_n$$

Find the two possible values of r.

2.08 — Question 55. The first three terms of a geometric series are given by $\frac{5}{k}$, $\frac{1}{k^4}$ and $\frac{k}{80}$, where k is a positive constant. Show that the sum to infinity can be written in the form $ak + b$, where a and b are rational numbers and k is a value to be found.

2.08 — Question 56. The first four terms of a geometric series are:

$$10 + \frac{10}{2k} + \frac{10}{(2k)^2} + \frac{10}{(2k)^3} + \cdots$$

where k is a positive constant. It is given that the series has a sum to infinity. State the possible values of k.

2.08 — Question 57. Find the value of

$$\sum_{n=1}^{\infty} (\frac{1}{2} \ln e^2 - 2 \ln e^{1/4})^n$$

2.08 — Question 58. Find the exact value of

$$\sum_{n=1}^{\infty}(\sin 45^\circ \cos 30^\circ)^n$$

2.08 — Question 59. Find the exact value of

$$\sum_{n=1}^{\infty}(\frac{\sin(90+180n)^\circ}{5})^n$$

2.08 — Question 60. Find the exact value of

$$\sum_{n=1}^{\infty}(\frac{\sqrt{5}}{2}-\frac{\sqrt{3}}{3})^n$$

2.08 — Question 61. A geometric sequence has nth term $u_n = \frac{100}{9} \times (\frac{3}{5})^{n-1}$. Solve the inequality

$$\sum_{n=1}^{\infty} u_n - \sum_{n=1}^{k} u_n > 0.09$$

where k is a constant.

2.08 — Question 62. A geometric sequence has first term a and common ratio r. Show that S_n, the sum of the first n terms, is $\frac{a(r^n-1)}{r-1}$.

2.08 — Question 63. A geometric sequence has nth term $u_n = 8000 \times (\frac{1}{2})^{n-1}$. Solve the inequality

$$\sum_{n=1}^{\infty} u_n - \sum_{n=1}^{k} u_n < 5000$$

where k is a constant.

2.08 — Question 64. A geometric sequence has nth term $u_n = k \times 0.5^{n-1}$, where k is a positive constant.

a) Find the value of the 5th term in terms of k

b) Show that the sum to infinity is $2k$

c) Explain how the sum to infinity would be affected if the nth term of the sequence was $v_n = 0.5 + k \times 0.5^{n-1}$

2.08 — Question 65. Jenny is an art student. She decides to paint a series of thin concentric circles on a canvas, with the first being the largest, having a radius of 40 cm, and the next one always has a radius that is 98% the length of the previous one. The total length of all the circumferences of the circles on the canvas will not exceed k cm. Find the value of k.

2.08 — Question 66. A geometric sequence has first term 108 and common ratio $\frac{3}{8}$. Show that the nth term for the sequence can be written as

$$\frac{3^{n+2}}{2^{3n-5}}$$

2.08 — Question 67. A geometric sequence has first term 400 and common ratio $\frac{5}{4}$. Show that the nth term for the sequence can be written as

$$\frac{5^{n+1}}{2^{2n-6}}$$

2.08 — Question 68. A geometric sequence has first term $(2k^2 + 5k - 6)$ and second term $(2k + 5 - \frac{6}{k})$, where k is an irrational constant. It is given that the sum to infinity is 3. Find the value of k.

2.08 — Question 69. It is given that

$$\sum_{n=1}^{\infty}(\frac{k+1}{2k+3})^n = 2$$

Find the value of

$$\sum_{n=1}^{6}(\frac{k+1}{2k+3})^n$$

2.08 — Question 70. Prerequisite knowledge for Q70–Q72: Recurrence Relations & Arithmetic Sequences Determine if any of these sequences are arithmetic, geometric, or neither:

a) $u_{n+1} = u_n + 3,\ u_1 = 5$

b) $u_{n+1} = 2u_n,\ u_1 = 4$

c) $u_{n+1} = 2u_n + 3,\ u_1 = 6$

2.08 — Question 71. It is given that u_n is an arithmetic sequence with first term 3 and common difference 4. It is also given that v_n is a geometric sequence with first term 2 and common ratio 3.

a) Determine whether the following sequence is arithmetic, geometric or neither:

$$(u_1 + v_1), (u_2 + v_2), (u_3 + v_3), \ldots, (u_n + v_n)$$

b) Find the exact value of

$$\sum_{n=1}^{8} (u_n + v_n)$$

2.08 — Question 72. An arithmetic progression has first term 10 and common difference 700, and a geometric progression has first term 5 and common ratio 3. The kth term for both sequences is the first time when the geometric term is greater than the arithmetic term.

a) Show that $3^k + 414 > 420k$

b) Determine the value of k

2.09 — Question 1. For each of the following, expand up to and including the x^3 term, stating in each case the values of x for which the expansions are valid:

a) $(1+x)^{\frac{1}{4}}$
b) $(1+2x)^{-1}$
c) $\sqrt{1-3x}$
d) $(2+x)^{-3}$
e) $(9+3x)^{\frac{1}{2}}$
f) $\sqrt{16-36x}$
g) $\sqrt[3]{8-2x}$
h) $\frac{1}{\sqrt{3+2x}}$

2.09 — Question 2. Write down the coefficient of the x^4 term in each of the following expansions:

a) $(1+3x)^{\frac{1}{4}}$

b) $(1-2x)^{\frac{2}{3}}$

c) $(1+\frac{x}{2})^{-\frac{3}{4}}$

2.09 — Question 3.

a) Find the first 3 terms in the expansion of $\frac{1}{\sqrt{9+x}}$

b) Hence find the first 3 terms in the expansion of $\frac{1}{\sqrt{9-x^2}}$

2.09 — Question 4.

a) Find the expansion of $(1+x)^{-\frac{1}{2}}$ up to and including the x^2 term.

b) Hence find the first 3 terms in the expansion of

$$(1+\tfrac{2}{5}x)^{-\frac{1}{2}} \quad \text{and} \quad \sqrt{\frac{10}{5+2x}}$$

2.09 — Question 5.

a) Find the first 3 terms in the expansion of $\sqrt{9-x}$

b) Hence find an approximation to $\sqrt{8}$

2.09 — Question 6.

a) Expand $(1+5x)^{\frac{1}{3}}$ up to and including the term in x^3.

b) Hence find an approximation to the value of $\sqrt[3]{1.1125}$ to 4 decimal places.

2.09 — Question 7. By finding the first 3 terms in the expansion of $(2-3x)^{\frac{1}{2}}$ use $x = \frac{1}{10}$ to find an approximation for $\sqrt{170}$ giving your answer in the form $p\sqrt{2}$ where p is a value to be found.

2.09 — Question 8.

a) Find the first 3 terms in the expansions of $(1+3x)^{\frac{1}{2}}$ and $(1+x)^{-\frac{1}{2}}$

b) Hence, show that

$$\sqrt{\frac{1+3x}{1+x}} \approx 1 + x - \frac{3}{2}x^2$$

c) State the values of x for which the expansion is valid.

2.09 — Question 9.

a) Find the first 3 terms in the expansion of $(1-2x)^{\frac{1}{2}}$

b) Hence, show that

$$\frac{\sqrt{1-2x}}{1+x} \approx 1 - 2x + \frac{3}{2}x^2$$

c) State the values of x for which the expansion is valid.

2.09 — Question 10. Given the binomial expansion

$$(1+qx)^p = 1 - 2x + 6x^2 + \dots$$

where p and q are constants.

a) Find the values of p and q.

b) Hence state the values of x for which the expansion is valid.

2.09 — Question 11. Given that

$$\sqrt[3]{8-x} \approx 2 - \frac{1}{12}x + kx^2 + \dots$$

where k is a constant.

a) Find the value of k.

b) By using $x = 1$ in the expansion, find an approximation to the value of $\sqrt[3]{7}$.

2.09 — Question 12.

a) By using $x = \frac{1}{8}$ in the first three terms in the expansion of $\sqrt{3-8x}$, find an approximation for the value of $\sqrt{2}$.

b) Explain whether your approximation is an under- or over-estimate.

2.09 — Question 13.

a) Find the terms up to and including the x^2 term in the expansion of $\frac{1}{\sqrt{2-x}}$

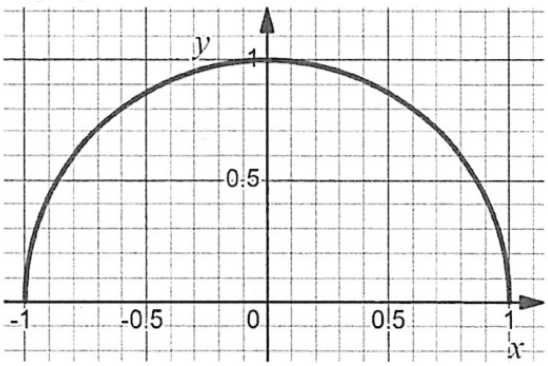
Jenna considers using the values $x = -3$, $x = -\frac{6}{5}$ and $x = \frac{1}{5}$.

b) Explain how she used the values to find $\sqrt{5}$.

c) Which of the three values should not be used in the expansion?

2.09 — Question 14. By considering the first three terms in the expansion of $\frac{1}{\sqrt{7-x}}$, find an approximation for the value of $\sqrt{3}$.

2.09 — Question 15. The graph below shows $y = \sqrt{1-x^2}$.



By considering the graph of $1 + \frac{1}{2}x^2 + \ldots$ explain why this is not a correct expansion of $\sqrt{1-x^2}$.

2.09 — Question 16. Find the values of a and b given

$$\frac{a+bx}{\sqrt{9-x}} = 1 - \frac{11}{18}x - \frac{7}{216}x^2$$

where a and b are constants.

2.09 — Question 17.

a) Find the first 4 terms in the expansion of $(1+px)^{\frac{1}{2}}$ where p is a constant

Given for the constant term q

$$(1+qx)(1+px)^{\frac{1}{2}} = 1 - 4x - 14x^2 + 16x^3 - \ldots$$

b) find the values of p and q

2.09 — Question 18.

a) Find the first 4 terms in the expansion of $(1-x)^{-2}$

b) Hence find $2+3x+4x^2+5x^3+\ldots$ in the form $\frac{a-x}{(b-x)^2}$ where a and b are integers to be found.

2.09 — Question 19. Prerequisite knowledge for Q19–Q21: Partial Fractions It is known that

$$\frac{2-5x}{(x-1)(2x-1)}, \qquad x \neq 1,\ x \neq \frac{1}{2}$$

can be expressed in the form $\frac{A}{x-1}+\frac{B}{2x-1}$.

a) Find the values of A and B.

b) Hence find the first three terms in the expansion of $\frac{2-5x}{(x-1)(2x-1)}$.

2.09 — Question 20. By expressing

$$\frac{11x+5}{(5x+1)(x+3)}, \qquad x \neq -\frac{1}{5},\ x \neq -\frac{1}{3}$$

in the form $\frac{A}{5x+1} + \frac{B}{x+3}$

a) Find the first 3 terms in the binomial expansion of $\frac{11x+5}{(5x+1)(x+3)}$.

b) State the values of x for which the expansion is valid.

2.09 — Question 21. By expressing

$$\frac{9x^2-5x+2}{(x+1)(3x+1)^2}, \qquad x \neq -1,\ x \neq -\frac{1}{3}$$

as partial fractions,

a) Find the first 3 terms in the binomial expansion of $\frac{9x^2-5x+2}{(x+1)(3x+1)^2}$.

b) State the values of x for which the expansion is valid.

2.09 — Question 22. Prerequisite knowledge for Q22: Integration of Polynomials

a) Find the first three terms in the expansion of $\frac{1}{\sqrt{9-x}}$

b) Hence find an approximation for $\int_0^1 \frac{1}{\sqrt{9-x^2}}\,dx$

2.09 — Question 23. Prerequisite knowledge for Q23: Radians A 1 radian sector of a circle has area $(x+2)$ units^2. Show that for small values of x, the radius is given by

$$r \approx 2 + px - qx^2$$

where p and q are constants to be found.